

Vascular smooth muscle cells in atherosclerosis

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Abstract | Vascular smooth muscle cells (VSMCs) are a major cell type present at all stages of an atherosclerotic plaque. According to the ‘response to injury’ and ‘vulnerable plaque’ hypotheses, contractile VSMCs recruited from the media undergo phenotypic conversion to proliferative synthetic cells that generate extracellular matrix to form the fibrous cap and hence stabilize plaques. However, lineage-tracing studies have highlighted flaws in the interpretation of former studies, revealing that these studies had underestimated both the content and functions of VSMCs in plaques and have thus challenged our view on the role of VSMCs in atherosclerosis. VSMCs are more plastic than previously recognized and can adopt alternative phenotypes, including phenotypes resembling foam cells, macrophages, mesenchymal stem cells and osteochondrogenic cells, which could contribute both positively and negatively to disease progression. In this Review, we present the evidence for VSMC plasticity and summarize the roles of VSMCs and VSMC-derived cells in atherosclerotic plaque development and progression. Correct attribution and spatiotemporal resolution of clinically beneficial and detrimental processes will underpin the success of any therapeutic intervention aimed at VSMCs and their derivatives.

Atherosclerosis is the formation of plaques containing lipids, cells, debris and scar tissue in the intima of arteries. As the main pathological process underlying myocardial infarction, angina, heart failure and stroke, atherosclerosis has been the leading cause of morbidity and mortality in the Western world for over half a century and is now the top cause of death globally¹. The prominent role of vascular smooth muscle cells (VSMCs) in atherosclerosis was established in the 1960s, as soon as electron microscopy made the identification of smooth-muscle-like cells in the media of normal arteries possible². These studies also ascertained that the majority of cells in atherosclerotic plaques had characteristics of VSMCs but with altered phenotypes^{3–5}. However, the perception of how VSMCs contribute to atherosclerotic plaque development, remodelling and stabilization has changed substantially over the past 50 years (BOX 1), and studies published in the past decade have questioned long-standing assumptions about the identity of cells in atherosclerotic plaques, demanding a re-evaluation of the role of VSMCs in atherosclerosis. In this Review, we aim to redress outdated perceptions concerning the role of VSMCs in atherosclerosis by critically evaluating the studies on the origin and fate of VSMCs in atherosclerosis and providing a holistic account of the evidence for the role of VSMCs as the disease progresses.

Identification of VSMCs

VSMCs are defined on the basis of anatomical localization (within the vasculature) and functionality. In healthy arteries, VSMCs are located in the medial layer, where they are responsible for arterial contraction and production of extracellular matrix (ECM) and have important roles in compliance and elastic recoil of the artery in response to changing haemodynamic conditions. VSMC functions are key determinants of the properties of vessels throughout the arterial tree; VSMC-derived elastin is crucial for elastic recoil in large elastic arteries (such as the aorta), and VSMC contraction is largely responsible for modulating arterial diameter in muscular arteries and arterioles (the latter being of great importance to systemic arterial resistance). Functionality is usually inferred from a combination of characteristics, including morphology and expression of ‘VSMC-specific’, function-associated markers (which are typically proteins and glycosaminoglycans). In healthy arteries, VSMCs have a fusiform shape, express contractile proteins (including α -smooth muscle actin (α SMA) and smooth muscle myosin heavy chain (SMMHC; also known as myosin 11 (MYH11)), which are organized into myofilaments) and secrete ECM macromolecules, including elastins, collagens and proteoglycans. Most studies to date have relied on these markers^{6–9} or on gene expression profiles¹⁰ for the

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<https://doi.org/10.1038/s41569-019-0227-9>

Key points

- Vascular smooth muscle cells (VSMCs) and VSMC-derived cells are a major source of plaque cells and extracellular matrix at all stages of atherosclerosis.
- VSMCs contribute to many different plaque cell phenotypes, including extracellular-matrix-producing cells of the fibrous cap, macrophage-like cells, foam cells, mesenchymal stem-cell-like cells and osteochondrogenic cells.
- Progress has been made in identifying the source of VSMCs and VSMC-derived cells in atherosclerotic plaques, highlighting the importance of the developmental origin, clonal expansion and phenotype switching of VSMCs in atherosclerosis.

identification of VSMCs. However, as a necessary corollary of their role in tissue homeostasis and repair, VSMCs have considerable phenotypic plasticity in atherosclerosis, in response to injury, and in culture in vitro, which is often accompanied by marked changes in cell morphology and expression of 'VSMC-specific' markers. Therefore, defining the cell type on the basis of functionality or 'specific' markers as a proxy for cell identification is problematic and has confounded studies on the true extent of the role of VSMCs in atherosclerosis¹¹.

Developments in genetic engineering have enabled specific labelling of VSMCs in mice, making fate mapping and lineage tracing of VSMCs possible. For example, inducible VSMC-labelling systems, such as a tamoxifen-inducible recombinase driven by promoters of 'VSMC-specific' genes¹² (typically *MYH11* (REF.¹³) or *TAGLN*^{14–16}) combined with reporter proteins^{17,18}, facilitate specific and stable labelling of VSMCs at baseline and enable unambiguous tracing of VSMCs and VSMC-derived cells during atherogenesis, even when VSMC characteristics might otherwise be lost or gained^{11,17–24}. This elegant approach has led to important advances in our understanding of the functional consequences of developmental origin, plasticity, clonality and, ultimately, fate of VSMCs in plaques, providing evidence for a more complex and prominent role for VSMCs and VSMC-derived cells in atherosclerosis than previously thought.

Origin of VSMCs

VSMCs are derived from multiple distinct progenitor cells in embryogenesis (BOX 2), with little or no mixing between different lineages^{25–27}, resulting in anatomical segmentation across the arterial tree. Furthermore, evidence indicates the presence of positional identity among VSMCs along the anterior–posterior, dorso–ventral and right–left axes of the embryo^{28–30}. Embryonic lineage can have important functional consequences; for example, VSMCs show lineage-dependent responses to important signalling pathways such as transforming growth factor- β (TGF β)^{31,32}, platelet-derived growth factor (PDGF)³³, myocardin-related transcription factor B (MRTF-B)^{34,35}, nuclear factor- κ B (NF- κ B)³⁶ and angiotensin II³⁷. These findings exemplify a fundamental limitation in defining VSMCs on the basis of 'VSMC-specific', function-associated markers, which can be similarly expressed in all VSMC lineages (potentially evoked through different pathways that converge on the same set of 'VSMC-specific' genes, as detailed in BOX 3), whereas different VSMC lineages can have distinct functional characteristics.

Lineage-tracing studies have unambiguously demonstrated that VSMCs contribute substantially to plaque formation in murine models of atherosclerosis, generating 30–70% of all plaque cells^{11,18–20,22,23}. In particular, most α SMA⁺ cells within the fibrous cap are positive for VSMC-lineage label^{38–40}, refuting earlier ideas^{41,42} that bone-marrow-derived cells generate α SMA⁺ cells. VSMC-derived cells that express mesenchymal stem cell markers, in particular stem cell antigen 1 (SCA1; also known as LY6A), have also been identified in the healthy artery media⁴³ and in atherosclerotic plaques^{11,43} and might represent a plastic intermediate population that is readily responsive to inflammation and capable of generating contractile or phenotypically switched VSMCs⁴³. However, these studies do not rule out a contribution from other sources of progenitor cells to plaque VSMCs (BOX 2).

Evidence for clonality (discussed below) of VSMCs and VSMC-derived cells in plaques indicates that the majority of plaque cells are derived from recruitment and proliferation of local VSMCs, and the anatomical localization of different developmental origins of VSMCs (and perhaps other cell types, such as pericytes and endothelial cells) might contribute to the anatomical localization of atherosclerosis susceptibility⁴⁴. This idea is supported by the finding that segments of aorta from atherosclerosis-prone and atherosclerosis-resistant regions maintain their atherosclerosis susceptibility after transplantation to alternative sites⁴⁵. Definitive evidence of similar anatomical segmentation of VSMC populations in humans is currently lacking, but the concept is supported in part by studies showing that human arteries are composed of clonal patches of VSMCs^{46–48}. Furthermore, advances in the understanding of the development of different VSMC lineages in vivo have led to the generation of VSMCs from stem cells in vitro⁴⁹, which will facilitate better disease modelling in human cells in vitro⁵⁰.

Plasticity of VSMCs

VSMCs have a fully functional, differentiated phenotype in healthy vessels, yet retain remarkable plasticity. De-differentiation, modulation or phenotype switching of VSMCs is characterized by reduced myofilament density and lower expression of contractile proteins. De-differentiated VSMCs have higher expression of ECM components and ECM-remodelling enzymes, have increased levels of secretory organelles and express pro-inflammatory cytokines⁵¹. Consequently, phenotypically switched VSMCs are often referred to as 'synthetic', whereas VSMCs expressing high levels of contractile proteins are generally described as 'contractile' (although these definitions imply explicit functional changes that are usually only inferred and very rarely quantified). Activation of VSMC proliferation and migration has also been associated with the synthetic, de-differentiated state, but coordinated regulation of these processes has not been documented, and mitotic VSMCs with high levels of contractile proteins have been observed^{52,53}.

Phenotypic switching is a reversible process, at least in the early stages. For example, a transient loss of contractile protein expression is observed after vascular

Lineage tracing

Technique that enables the fate of labelled cells to be traced in vivo to allow the identification of progeny cells.

Mesenchymal stem cell

Multipotent stromal cell.

Phenotype switching

Process by which vascular smooth muscle cells (VSMCs) alter their phenotype, often inferred through decreased expression of VSMC-specific genes encoding contractile proteins and/or increased expression of markers typical of synthetic VSMCs or other cell types.

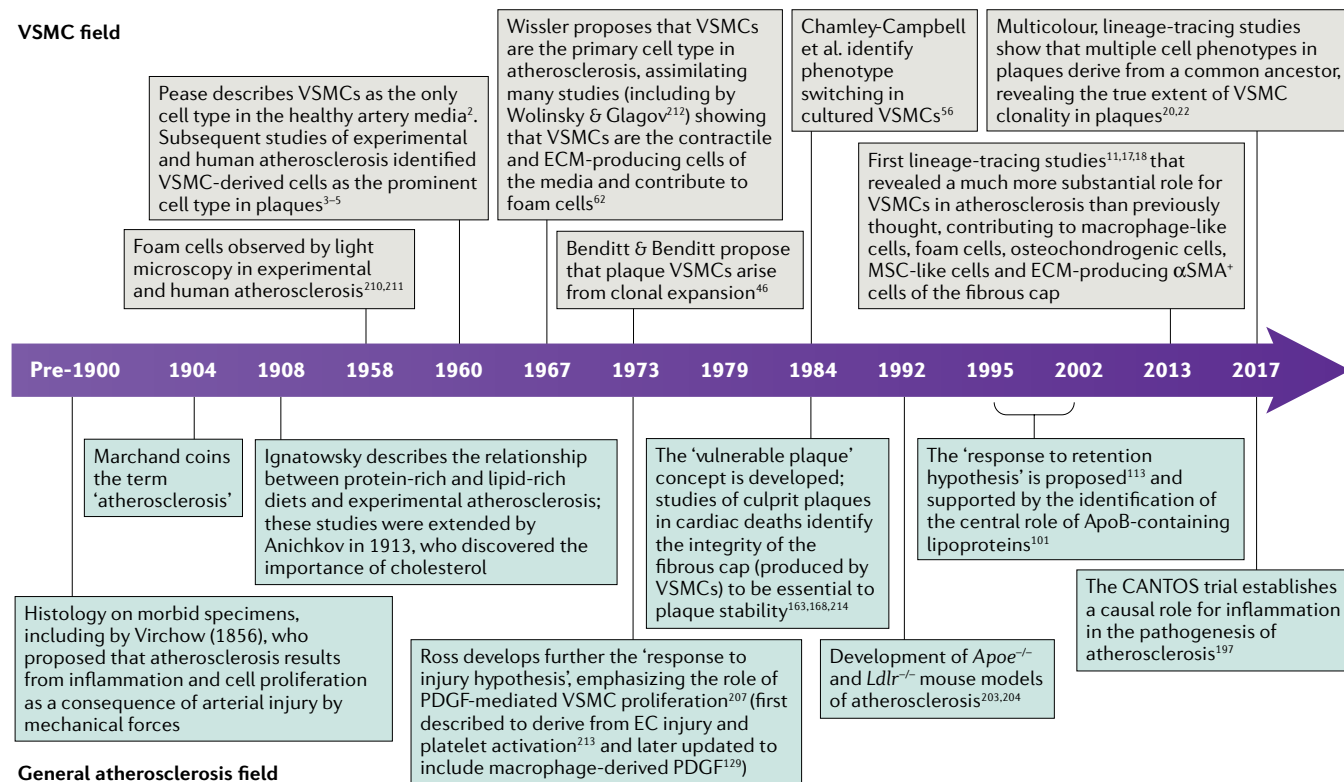
Box 1 | Historical perspective on VSMCs in atherosclerosis

The development of antibodies for 'vascular smooth muscle cell (VSMC)-specific', function-associated markers, such as α -smooth muscle actin (α SMA)^{6–9}, greatly facilitated immunohistological studies of VSMCs in atherosclerotic plaques in animal models^{203,204} and humans^{98,103}. These studies, alongside data on in vitro culture models⁵⁶ and models of arterial injury, such as balloon angioplasty, revealed that VSMCs have great phenotypic plasticity and undergo 'phenotypic switching' from contractile to proliferative synthetic phenotypes^{205–207}. Phenotype switching and proliferation of VSMCs in response to arterial injury and lipid infiltration into the arterial wall were considered the main pathological processes underlying atherosclerotic plaque development²⁰⁷.

Studies in the 1990s characterized the role of VSMC proliferation, migration, apoptosis and phenotype switching in atherogenesis²⁰⁸ and revealed that VSMCs can give rise to foam cells^{4,5,102} and osteochondrogenic cells¹⁵⁴. However, detailed post-mortem analyses of culprit plaques in sudden cardiac death established that the integrity

of the fibrous cap, comprising mostly α SMA⁺ cells and associated extracellular matrix (ECM), is critical to stabilize and protect atherosclerotic plaques from rupture, a major cause of the clinical sequelae of atherosclerosis^{142,163,168}. These studies also highlighted the role of immune cells, particularly macrophages, and inflammation as the main drivers of plaque development¹⁶⁹. Thus, the prevailing model has been that VSMCs contribute to the cellularity and inflammation of the developing plaque but have a predominantly beneficial role in its stabilization through the generation of the fibrous cap²⁰⁹ (see the figure).

In the past decade, studies applying fate-mapping and lineage-tracing techniques have revealed the limitations of relying on 'VSMC-specific', function-associated markers to infer VSMC identity^{11,17,18}. Furthermore, they have exposed the extent to which use of these markers can lead to false-negative and false-positive identification of VSMCs, as well as the oversimplification of VSMC heterogeneity and functions in atherosclerotic plaques.



ApoB, apolipoprotein B; EC, endothelial cell; MSC, mesenchymal stem cell; PDGF, platelet-derived growth factor.

injury, followed by re-establishment of the contractile phenotype after vessel repair⁵⁴. VSMCs with phenotypes across a spectrum ranging from contractile to synthetic states have also been observed both in vivo⁵³ and in VSMC cultures in vitro^{55,56}, illustrating that phenotypic switching is not a binary process. VSMC heterogeneity in morphology and gene expression^{43,57} is also seen in healthy vessels, including detection of rare, atypical VSMCs positive for SCA1 and expressing genes associated with phenotypic switching⁴³. At the molecular level, the VSMC phenotype is governed by transcription factors — including myocardin–serum response factor (SRF)⁵⁸ and Krüppel-like factor 4 (KLF4)¹¹, which integrate input from the environment (including growth factors, cytokines and lipid mediators, and contact with

the ECM and other cells) — and is regulated at multiple levels, including by epigenetic mechanisms (BOX 3).

Lineage-tracing studies have revealed that VSMCs have greater than anticipated plasticity in atherosclerosis (TABLE 1). A large proportion of reporter-expressing, VSMC-derived cells within atherosclerotic plaques do not have detectable levels of the contractile smooth muscle cell marker α SMA^{11,20}. Instead, some reporter-expressing cells in the plaque were positive for lysosome-associated membrane glycoprotein 2 (LAMP2; also known as CD107b and MAC3)²⁰, galectin 3 (GAL3; also known as MAC2)¹¹ and CD68 (REF. 17), which are markers that have been previously used to study macrophages in atherosclerosis. Stimulation of VSMCs in vitro with cholesterol similarly induces expression of macrophage-associated

Box 2 | Embryonic origins of VSMCs and sources of VSMC progenitors in adults

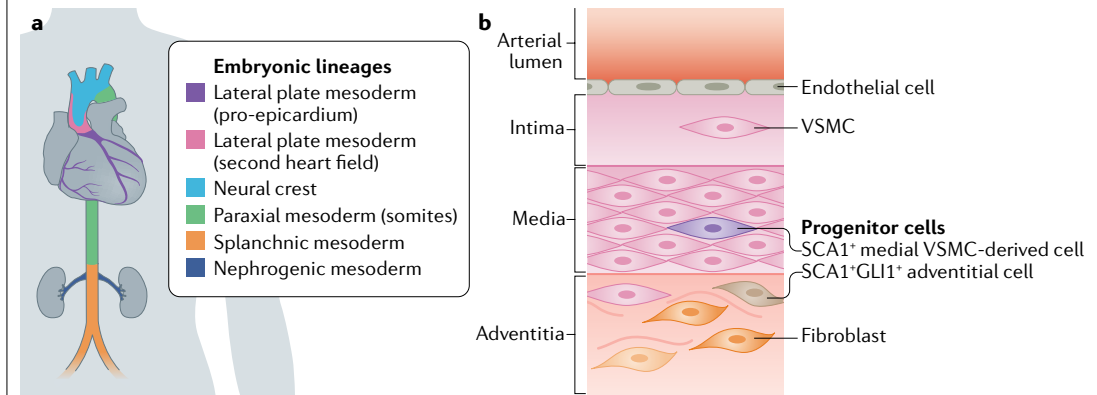
VSMC progenitor cells in embryogenesis

During embryonic development, medial vascular smooth muscle cells (VSMCs) — and in some instances pericytes²¹⁵ — arise from local progenitor cells, of which there are multiple distinct cell lineages distributed across the arterial tree (see the figure, part a). In mice, more than eight distinct progenitor cell populations have been identified^{44,216,217}. The aortic root and outer medial layers of the ascending aorta derive from the second heart field^{26,28}; the inner medial layer of the ascending aorta, aortic arch, ductus arteriosus, innominate and right subclavian arteries and right and left common carotid arteries derive from the neural crest²⁵; the descending aorta derives from paraxial (somatic) mesoderm²¹⁸; and the coronary arteries derive from pro-epicardium, which derives from lateral plate mesoderm²¹⁹.

VSMC progenitor cells in adults

Potential VSMC progenitor populations have also been identified in the media in the adult mouse, including VSMC-derived cells expressing stem cell antigen 1 (SCA1) and other mesenchymal stem cell markers^{11,43} (see the figure, part b). These cells might be an intermediate population derived from phenotypic switching, which can give rise to different VSMC-derived cell phenotypes⁴³. Other potential VSMC progenitor cells include a population of adventitial cells located close to the medial boundary, which express mesenchymal stem cell markers (such as SCA1) and are responsive to sonic hedgehog signalling (are GLI1⁺^{27,220–222}, and pericytes^{223,224}, which are VSMC-like cells of the microvasculature.

Importantly, studies have shown that progenitor cells with distinct origins can achieve a common VSMC fate with respect to expression of 'VSMC-specific', function-associated markers (through pathways discussed in BOX 3) but are nonetheless distinct with respect to other functional characteristics, such as responses to growth factors.



genes^{55,59} and promotes a phagocytic phenotype¹¹. VSMC-derived cells in human atherosclerotic plaques were also found to express CD68 (REF.¹¹), consistent with previous studies detecting cells co-expressing CD68 and α SMA in human atherosclerotic plaques^{60,61}. These results support the hypothesis proposed by Wissler in 1968 that at least a subset of foam cells in atherosclerotic plaques are derived from VSMCs⁶². This concept should be considered when interpreting studies on macrophage function that rely only on marker expression. Similarly, VSMCs have been proposed to generate osteochondrogenic cells and mesenchymal stem cell-like cells in atherosclerotic plaques on the basis of the expression of mineralizing ECM proteins^{63,64} and SCA1 and endoglin¹¹, respectively. The expanded plasticity of VSMCs in atherosclerosis was confirmed by transcriptional profiling of individual VSMC-lineage cells in mouse atherosclerotic plaques, revealing subpopulations of cells expressing SCA1, CD68 and the chondrocyte-related proteins transcription factor SOX9 and chondroadherin⁴³.

Clonality of VSMCs

The combination of multicolour recombination markers (such as the confetti or rainbow system^{18,22}) with genetic lineage tracing of VSMCs has demonstrated that, surprisingly, VSMC-derived cells in mouse atherosclerotic

plaques are generated by clonal expansion of few cells within the vessel wall^{17,20,22,23}. By contrast, most medial cells do not contribute to atherosclerotic plaque formation in mice and the role of VSMC migration independent of proliferation in this process is limited²⁰. Indeed, phenotypically distinct VSMC-derived cells in mouse atherosclerotic plaques are generated from a common 'ancestor'. Observations of atherosclerotic plaques at different time points suggest that, in mice, VSMCs first generate the fibrous cap, followed by phenotype switching in the lesion core²³, but this observation remains to be tested experimentally.

The molecular mechanisms underlying clonality are yet to be established, but macrophage-secreted factors have been implicated in this process. For example, bone-marrow transplantation from β 3-integrin-deficient mice into *Apoe*^{-/-} mice results in polyclonal VSMCs and VSMC-derived cells in the atherosclerotic plaque²³, and conditioned media from cultured macrophages deficient in β 3 integrin is more mitogenic to VSMCs than conditioned media from wild-type macrophages²³. VSMCs in the early-stage fibrous cap are highly proliferative and express α SMA, SMMHC and, importantly, platelet-derived growth factor receptor- β (PDGFR β)²³, akin to the primed PDGFR β ⁺ VSMC progenitors reported in models of hypoxia-induced pulmonary hypertension, which clonally expand in a

Foam cells

Lipid-laden cells with a foamy appearance.

Osteochondrogenic cells

Cells capable of generating osteocytes and/or chondrocytes.

Clonal expansion

Proliferation of a single or limited number of ancestral cells.

PDGF-dependent manner^{65,66}. This observation highlights a potential role for PDGF signalling in clonal expansion of VSMCs and demonstrates that the study of VSMC clonal expansion in other vascular conditions^{20,65,67} might be relevant for further mechanistic dissection in atherosclerosis.

The small number of VSMCs contributing to atherosclerotic lesion formation raises the question of whether disease-associated proliferation results from activation of specific cells that are primed to respond to injury (discussed previously⁶⁸). Supporting this idea, transcriptional profiling of VSMCs from healthy blood vessels revealed substantial heterogeneity in the expression of genes associated with vascular disease, suggesting

the existence of VSMC subtypes^{43,57}. Alternatively, clonality might rely on selection of VSMCs with equal plasticity, on the basis of location (for example, proximal to breaks in the internal elastic lamina and/or to mitogenic signals) or differential capacity for survival or senescence (see below). A role for pathways of lateral inhibition, which are common in development, has also been proposed²². Importantly, these possibilities are not mutually exclusive, and the underlying mechanism is likely to be genetic (somatic mutations) and/or epigenetic changes in the expanded VSMCs relative to non-expanded VSMCs.

Evidence that somatic mutations underlie clonal expansion in both malignant and non-malignant tissues

Box 3 | Molecular mechanisms underlying VSMC plasticity

Transcription factors

Myocardin. Myocardin (MYOCD) family proteins drive the expression of contraction-related genes⁵⁸. MYOCD is a cofactor for serum response factor (SRF), which binds to CArG-box elements within the promoters of contraction-related genes (see the figure). Most environmental cues and signalling pathways affecting vascular smooth muscle cell (VSMC) function alter the expression and/or activity of MYOCD^{225,226}.

KLF4. Krüppel-like factor 4 (KLF4) represses the expression of contraction-related genes through several mechanisms, including by binding to G/C repressor elements and by inhibiting SRF binding to CArG-boxes (see the figure). KLF4 inhibits proliferation; VSMC-specific deletion of DDIT3 leads to decreased VSMC proliferation through increased expression of KLF4 (REF.²²⁷). Importantly, VSMC phenotype switching is KLF4-dependent. KLF4 is required for induction of progenitor cells before clonal expansion of pulmonary VSMCs in hypoxia^{65,66}, and VSMC-specific deletion of KLF4 in *Apoe*^{-/-} mice results in reduced numbers of VSMC-derived, macrophage-marker and mesenchymal stem-cell-marker-positive cells in atherosclerotic plaques¹¹.

Extracellular stimuli

The VSMC contractile phenotype is promoted by transforming growth factor- β (TGF β) signalling, whereas platelet-derived growth factor (PDGF) induces KLF4 expression, VSMC proliferation and phenotypic switching to a synthetic phenotype (see the figure). Other growth factors, including WNT, also promote proliferation and migration of VSMCs. Pro-inflammatory cytokines (such as IL-1 and tumour necrosis factor (TNF)) alter the VSMC phenotype via nuclear factor- κ B (NF- κ B)-mediated and AP-1-mediated regulation of gene expression, including downregulation of MYOCD expression. Cholesterol-induced expression of macrophage-associated genes in VSMCs occurs via miR-143–miR-145, involves MYOCD and inflammatory signalling and is influenced by KLF4 (REFS^{59,228}).

Cell interactions

Extracellular matrix proteins and heparin alter the VSMC phenotype²²⁹. Notably, deletion of the gene encoding β 3 integrin in mice results in larger lesions and affects VSMC clonality in atherosclerosis²³. Differences in how cells communicate with the environment can also explain the documented effect of stretch and shear stress on VSMC phenotype²³⁰.

Epigenetic regulation

The reversibility of VSMC phenotypic switching indicates a cellular memory of the contractile state. Indeed, contraction-related genes retain the epigenetic modification histone H3 lysine 4 dimethylation (H3K4me2; which is generally associated with actively transcribed genes) after phenotypic switching¹⁸, and manipulation of DNA methylation and histone-modifying enzymes directly affects VSMC behaviour in murine models of vascular injury and atherosclerosis^{231–233}, whereas levels of epigenetic markers are altered in human atherosclerotic plaques²³⁴. Non-coding RNAs also control VSMC plasticity^{235,236}, evidenced by the effect of specific microRNAs and long non-coding RNAs on VSMC biology and function^{237,238}.

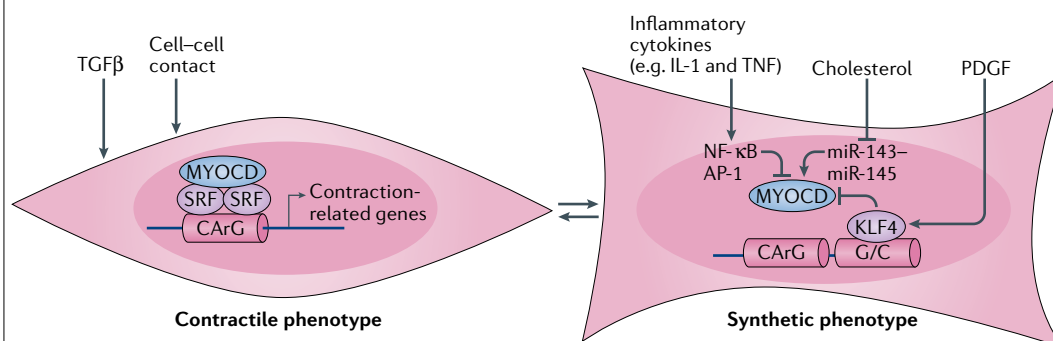


Table 1 | Lineage-tracing studies in atherosclerosis

Cell type traced	Tracing method ^a	Mouse model	Contribution of labelled cells to plaque?	Phenotype of labelled cells in plaques					Refs
				VSMCs	α SMA ⁺ cells	Macrophage-like cells	Osteochondrogenic cells	MSC-like cells	
VSMCs	Tagln-CreER ^{T2} ; R26R-lacZ or R26R-mT/mG or R26R-Confetti	<i>Apoe</i> ^{-/-} ; chow diet (52 weeks) or HFD (16 weeks)	Yes, clonal patches	Yes, α SMA ⁺	Yes	Yes, GAL3 ⁺ and CD68 ⁺ (62% and 54% of labelled cells, respectively)	NA	NA	17
	Myh11-CreER ^{T2} ; R26R-EYFP	<i>Apoe</i> ^{-/-} ; HFD 18 weeks	Yes	Yes, α SMA ⁺	Yes; >95% of labelled cells	NA	NA	NA	18
	Myh11-CreER ^{T2} ; R26R-EYFP	<i>Apoe</i> ^{-/-} ; HFD 18 weeks	Yes	Yes, 16% of labelled cells were α SMA ⁺	Yes, 12% of labelled cells were PDGFR β ⁺ , 32–51% of labelled cells were of unknown identity	Yes, 30% of labelled cells were GAL3 ⁺	NA	Yes, 7% of labelled cells were SCA1 ⁺	11
	Myh11-CreER ^{T2} ; R26R-Confetti	<i>Apoe</i> ^{-/-} ; HFD 16–19 weeks	Yes, 70% (40–90%) of plaque cells, oligoclonal patches	Yes, 30–100% of labelled cells were α SMA ⁺ ; 70–100% of α SMA ⁺ cells were labelled	Yes	Yes, 5–50% of labelled cells were LAMP2 ⁺ , 70% of LAMP2 ⁺ cells were labelled	NA	NA	20
	Myh11-CreER ^{T2} ; R26R-Confetti	PCSK9-D377Y AAV; HFD 12–36 weeks	Yes, oligoclonal contribution to cap and core	Yes, α SMA ⁺	Yes	Yes, Oil Red O ⁺ , no GAL3 ⁺ cells detected	Yes	Yes	22
	Myh11-CreER ^{T2} ; Brainbow	<i>Apoe</i> ^{-/-} ; HFD 5–12 weeks	Yes, monoclonal contribution to cap and core	Yes, α SMA ⁺	Yes	NA	NA	NA	23
	Myh11-CreER ^{T2} ; R26R-Confetti	<i>Apoe</i> ^{-/-} ; HFD 16–19 weeks	Yes, clonal patches	Yes, α SMA ⁺	Yes	Yes	Yes, SCA1 ⁺ (rare)	Yes, SCA1 ⁺ (rare)	43
Unknown	Chimaeras	<i>Apoe</i> ^{-/-} ; chow diet 10 months	Yes, oligoclonal patches in plaque cap	Yes, clonal, α SMA ⁺	NA	NA	NA	NA	22
TCF21 ⁺ cells (adventitial)	Tcf21-MerCreMer; R26R-tdTomato	<i>Apoe</i> ^{-/-} ; HDF 12 weeks, <i>Ldlr</i> ^{-/-} HFD 16–20 weeks	Yes	Yes, TAGLN ⁺	Yes, periostin ⁺	NA	NA	NA	239
Adventitial cells	Transplantation of cultured SCA1 ⁺ adventitial cells from SM-lacZ; <i>Apoe</i> ^{-/-} donor animals into <i>Apoe</i> ^{-/-} hosts	Vein graft	Yes	Yes, lacZ ⁺ cells in plaque	NA	NA	NA	NA	220
Adventitial MSCs	Gli1-CreER ^{T2} ; R26R-tdTomato	<i>Apoe</i> ^{-/-} with subtotal (5/6) nephrectomy and HFD 10–16 weeks	Yes	Yes, calponin ⁺ (20–80% of lineage-traced cells)	Yes	No CD68 ⁺ cells detected	Yes, calcium-tracer ⁺ RUNX2 ⁺ (10–25% of lineage-traced cells)	Yes, calcium-tracer ⁺ RUNX2 ⁺ (10–25% of lineage-traced cells)	221

Table 1 (cont.) | Lineage-tracing studies in atherosclerosis

Cell type traced	Tracing method ^a	Mouse model	Contribution of labelled cells to plaque?	Phenotype of labelled cells in plaques					Refs
				VSMCs	α SMA ⁺ cells	Macrophage-like cells	Osteochondrogenic cells	MSC-like cells	
BM-derived cells	BM from GFP ⁺ donor animals transplanted into GFP ⁺ hosts	<i>Apoe</i> ^{-/-} ; HFD 20–32 weeks	Yes, GAL3 ⁺ foam cells in plaque core	No	Yes	Yes, GAL3 ⁺	NA	NA	39
	BM from GFP ⁺ donor animals transplanted into GFP ⁺ hosts	Healing plaque (<i>Apoe</i> ^{-/-} mice with spontaneous or mechanically disrupted haemorrhagic plaque)	Yes, GAL3 ⁺ foam cells in plaque core	No	Yes	Yes, GAL3 ⁺	NA	NA	38
	BM from Myh11-Cre;R26R-lacZ; <i>Apoe</i> ^{-/-} donor animals transplanted into <i>Apoe</i> ^{-/-} hosts	<i>Apoe</i> ^{-/-} ; HFD 6–22 weeks	Yes, 0.7% of cells in advanced plaque were lacZ ⁺	Very rare (0.4% of plaque cells were α SMA ⁺ lacZ ⁺)	Very rare	NA	NA	NA	40
Endothelial cells	end.SclCreERT; R26R-EYFP	<i>Apoe</i> ^{-/-} ; HFD 8–30 weeks	Yes	Yes, low contribution (α SMA ⁺ or SMMHC ⁺) or TAGLN ⁺	Yes, 32–45% of FAP ⁺ fibroblasts were labelled	NA	NA	NA	240

α SMA, α -smooth muscle actin; AAV, adeno-associated virus; BM, bone marrow; FAP, prolyl endopeptidase FAP; GAL3, galectin 3; GFP, green fluorescent protein; HFD, high-fat diet; LAMP2, lysosome-associated membrane glycoprotein 2; MSC, mesenchymal stem cell; NA, not analysed; PDGFR β , platelet-derived growth factor receptor- β ; R26R, ROSA26 locus reporter; RUNX2, Runt-related transcription factor 2; SCA1, stem cell antigen 1; SMMHC, smooth muscle myosin heavy chain; TAGLN, transgelin; TCF21, transcription factor 21; VSMC, vascular smooth muscle cell. ^aTamoxifen-induced recombination before induction of atherosclerosis.

as a consequence of ageing is well documented⁶⁹. Indeed, the acquisition of a particular set of somatic mutations, linked to clonal expansion, in myeloid progenitor cells has been associated with increased risk of atherosclerosis⁷⁰. Therefore, similar mechanisms might underlie clonal expansion of VSMCs in atherosclerosis. Indeed, when clonal expansion of VSMCs was first described in atherosclerotic plaques, the process was likened to that in a smooth muscle cell tumour⁴⁶. Epigenetic changes might influence clonal expansion of VSMCs secondary to, or independently of, somatic mutations. Such changes might reflect differences in VSMC lineage, environmental stimuli or stochastic events.

Although lineage tracing has provided the most robust evidence yet for clonality of VSMCs in atherosclerotic plaques, the concept that most plaque VSMCs derive from clonal expansion, attributed to Benditt and Benditt⁴⁶, has long been discussed⁴⁷, particularly in the context of replicative senescence⁷¹.

VSMC senescence

Senescence is a protective mechanism that induces cell cycle arrest to prevent transmission of defects to progeny cells, particularly to stop malignant transformation^{72–74}. Replicative senescence occurs after repeated cell division, typically after telomere erosion or damage, whereas induced senescence arises after oncogene activation, mitochondrial deterioration, DNA damage

or oxidative stress. A persistent DNA damage response (DDR) is the most unified pathway leading to senescence, with sensing by the protein ataxia telangiectasia mutated (ATM) leading to p53 phosphorylation and upregulation of cell cycle inhibitors^{72–74}. The cyclin-dependent kinase (CDK) inhibitor 1 (also known as p21) drives initial cell cycle arrest, enabling the repair of moderate DNA damage and re-entry into the cell cycle. However, prolonged cell cycle arrest upregulates the CDK inhibitor 2A (also known as p16^{INK4A}), leading to dephosphorylation of retinoblastoma protein (pRb), causing permanent cell cycle arrest^{72–74}. With every somatic cell division, approximately 20 bp or more are lost from the telomere ends of the chromosomes. Therefore, repeated cell division leads to critical shortening and erosion of the telomeres and loss of the protective shelterin complex, which results in a persistent DDR that instigates senescence.

VSMC senescence in vivo is likely to be driven by multiple pathways, including DNA damage, mitochondrial deterioration and oxidative stress, which are all present during atherosclerosis. Loss of autophagy can also drive VSMC senescence⁷⁵. Replicative senescence is highly relevant in the context of atherosclerotic plaque VSMC clonality because the generation of all the VSMC-derived cells in advanced plaques by clonal expansion would probably cause replicative senescence. In keeping with this idea, the telomeres of VSMCs in

Shelterin complex
Multiprotein complex (including telomeric repeat-binding factor 2 (TRF2)) that binds to the repetitive sequences of telomeric DNA, protecting against DNA damage.

human atherosclerotic plaques are markedly shortened, which correlates with disease severity⁷⁶.

Most senescent cells develop altered secretory activities known as a senescence-associated secretory phenotype (SASP)^{77,78}. Cells with SASPs release pro-inflammatory cytokines (such as IL-6 and IL-1), chemokines (such as IL-8, CC-chemokine ligand 2 (CCL2; also known as MCP1) and CXC-chemokine ligand 1 (CXCL1; also known as GRO α)), growth factors (such as granulocyte colony-stimulating factor (G-CSF) and fibroblast growth factor 2 (FGF2)) and proteases (including matrix metalloproteinases (MMPs) and plasminogen activator inhibitor 1 (PAI1)), which confer diverse activities⁷⁸. IL-1 α is the key driver of the SASP^{79,80}, with upstream expression of IL-1 α controlled in part by ATM–ATR-mediated inhibition of the p62-directed degradation of the transcription factor GATA4 (REF.⁸¹) and/or by an mTORC1-dependent pathway⁸². In a physiological setting, SASPs act as a molecular beacon that recruit and instruct immune cells to remove senescent cells (senescent surveillance⁸³) before further mutation enables senescence bypass and, for example, re-initiation of tumour formation. However, senescent cells that are not cleared by phagocytes accumulate during ageing and disease (perhaps owing to a dysfunctional immune system or a suppressive milieu) and promote chronic inflammation that could drive atherosclerosis and/or worsen the outcome⁸⁴.

Although VSMC senescence occurs in human atherosclerotic plaques, proving the effects of senescent VSMCs is difficult and is hampered by technical difficulties in mouse models. For example, given that telomeres are approximately ten times longer in mice than in humans, studying mouse SASPs in vitro is problematic⁸⁵. Furthermore, detecting senescence with the classic markers p16^{INK4A} and senescence-associated β -galactosidase (SABG) is also notoriously difficult in mice, particularly when both markers are expressed by macrophages in atherosclerotic plaques. Two main experimental approaches have been used to study the effect of VSMC senescence in atherosclerosis: modulation of senescence induction via the DDR and clearance of naturally occurring senescent cells with ‘senolytics’. For example, VSMC-specific expression of a loss-of-function mutant of the shelterin subunit telomeric repeat-binding factor 2 (TRF2; also known as TERF2) in mice led to increased DNA damage and VSMC senescence, with bigger atherosclerotic plaques and necrotic cores, whereas expression of a gain-of-function TRF2 mutant produced opposite effects⁸⁶. Similarly, VSMCs that lack base excision repair activity have increased oxidative DNA damage and cell senescence and promote increased atherosclerotic plaque size in mice⁸⁷. By contrast, an intriguing study used electron microscopy to assess crystals in mouse atherosclerotic plaques proposed to be the product of X-galactosidase cleavage by SABG⁸⁴. This study reported that >50% of plaque cells were senescent, including VSMCs, macrophages and endothelial cells⁸⁴. Senescent cells appeared in aortic lesions after 9 days of high-fat diet, and both genetic and pharmacological elimination of p16^{INK4A}-positive cells reduced atherosclerotic plaque formation and progression⁸⁴. Although the cell types

that were senescent and removed by these treatments are unclear, this approach might open a new paradigm for atherosclerosis treatment.

VSMCs at different stages of disease

Histological studies of atherosclerotic plaques from human autopsy tissues have culminated in a scheme for classification of lesions that encapsulates the progression of atherosclerosis^{88,89}, and careful observations of atherosclerotic plaque composition from human autopsies and animal models have revealed that VSMCs are major contributors to plaque development at all stages (summarized in FIG. 1). However, the role of VSMCs and the effects of VSMC proliferation or loss differ according to the stage of atherogenesis.

Pre-atherosclerosis. Diffuse intimal thickenings (DITs) and intimal xanthomas (or fatty streaks) are considered pre-atherosclerotic plaques because they are common from birth^{90,91} and probably represent a physiological adaptation to blood flow⁹². However, the relationship between intimal xanthomas and atherosclerosis is controversial because, although these lesions localize to atherosclerosis-prone regions and some intimal xanthomas develop into atherosclerotic plaques, they are also found elsewhere in the arterial tree and sometimes regress^{93–95}. By contrast, DIT distribution in young individuals is similar to that of atherosclerotic plaques in later life^{90,96}, and DITs are widely considered the most likely precursor of atherosclerotic plaques⁸⁸.

Human DITs comprise VSMCs, proteoglycans and elastin and lack macrophages and thrombus^{88,91,92}. VSMCs in DITs exhibit clonality^{47,91} and are thought to originate from local medial VSMCs⁵⁷. However, the latter is difficult to prove because many lineage-tracing techniques (for example, reporter gene expression under the regulation of a lineage-specific promoter) are limited to animal models, and most mammals (including mice) do not develop DITs⁹⁷. VSMCs in DITs are heterogeneous, but most have more synthetic organelles (rough endoplasmic reticulum, ribosomes and mitochondria) than medial VSMCs⁹⁸, consistent with switching to a synthetic phenotype, which is supported by decreased expression of genes encoding contractile proteins⁹⁹ and increased generation of ECM components¹⁰⁰. VSMCs are thought to be the major source of ECM in DITs, which accounts for much of the increase in thickness of the intima, but, importantly for the progression to atherosclerosis, DITs are rich in proteoglycans, which are crucial for the retention of apolipoproteins¹⁰¹. Furthermore, VSMCs with a synthetic phenotype metabolize lipids differently to contractile VSMCs, in part through decreased expression of cholesterol esterase and reduced levels of the cholesterol efflux transporter ATP-binding cassette transporter ABCA1 (REFS^{60,102}), resulting in an increased tendency to transform into foam cells¹⁰³.

Early atherosclerosis. The first stage in atherosclerosis is pathological intimal thickening (PIT), which is characterized by the formation of extracellular lipid pools deep in the arterial intima, underlying abundant VSMCs and ECM^{88,89}. DITs can, but do not always, progress to

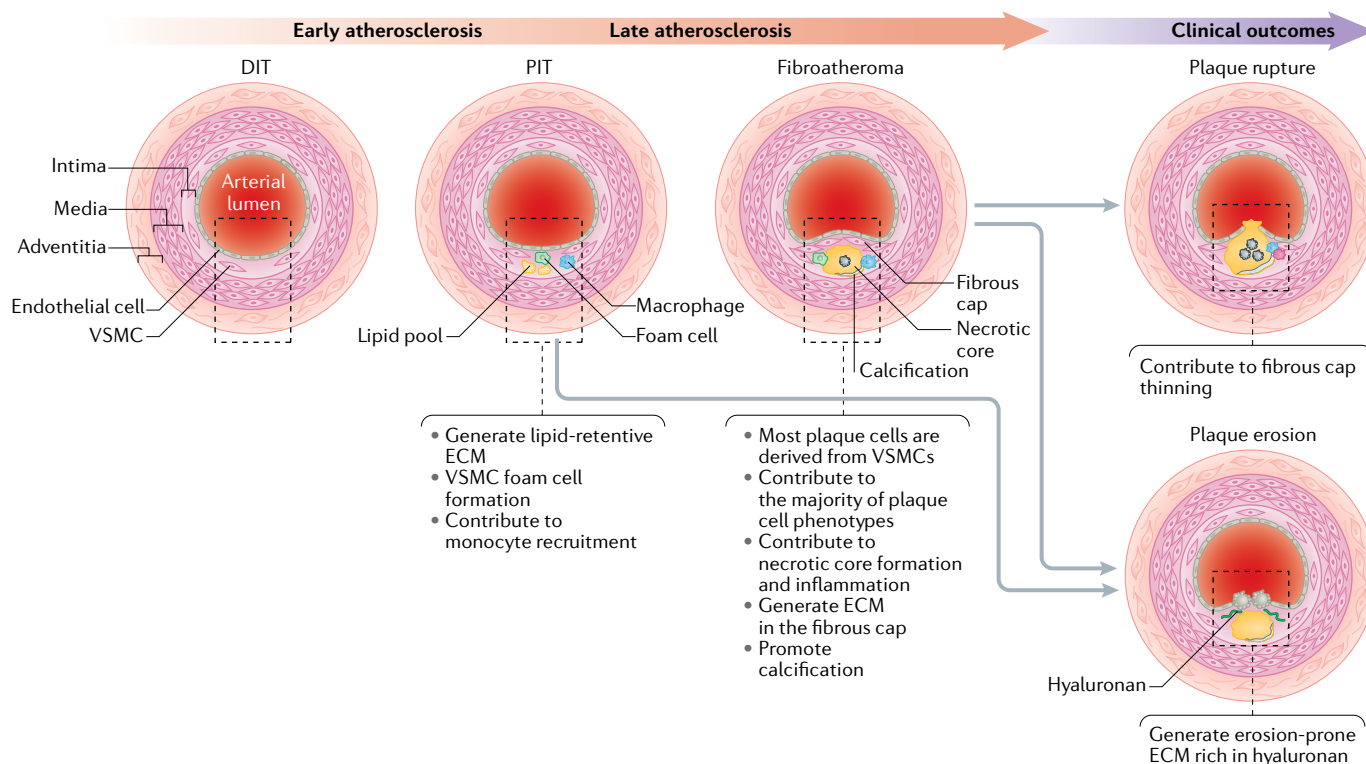


Fig. 1 | Overview of the role of VSMCs in atherosclerosis. Vascular smooth muscle cells (VSMCs) are a major source of plaque cells and extracellular matrix (ECM) at all stages of atherosclerosis and contribute to numerous processes throughout the disease. DIT, diffuse intimal thickening; PIT, pathological intimal thickening.

PITs¹⁰⁴ (FIG. 2). Progression from DIT to PIT is promoted through a complex interplay between retention and oxidation of lipids, induction of inflammation and VSMC proliferation, phenotype switching and death.

The lipid pools in PIT (which are distinct from the necrotic pools of more advanced plaques) comprise lipids (including free cholesterol) amidst an ECM rich in proteoglycans (notably biglycan, versican and perlecan) and glycosaminoglycans (including hyaluronan). As the predominant cell type present in DITs, intimal VSMCs are considered to be the most important source of ECM components, and this idea is supported by findings from the analysis of the secretome of VSMCs *in vitro*^{105–109}. The ECM has a central role in initiation of atherosclerosis, primarily through the interaction between the negatively charged side chains of proteoglycans (particularly chondroitin sulfate of biglycan and versican and heparin sulfate of perlecan¹¹⁰) with positively charged apolipoproteins (especially apolipoprotein B). This interaction leads to the retention of plasma-derived lipoproteins^{101,111}, as described in the ‘response to retention hypothesis’^{112,113}. Transgenic mice overexpressing biglycan in VSMCs show more lipid retention in the vessel wall and increased atherosclerosis compared with wild-type littermates¹¹⁴. Once retained in the intima, lipoproteins undergo modifications, including oxidation of LDL (oxLDL), which precedes the recruitment of macrophages¹¹⁵ and initiates the inflammatory response characteristic of atherosclerosis¹¹². Further evidence for this series of events was provided by a study assessing DIT and PIT, which showed that

extracellular lipid pools were deep in PIT lesions, colocalizing with α SMA⁺ cells, biglycan and versican but not with the more superficial (closer to the lumen) CD68⁺ cells (probably macrophages)¹¹⁶.

Progression from DIT to PIT is accompanied by loss of α SMA expression, which is likely to be due to a combination of phenotypic switching of VSMCs^{11,18,23} and loss of VSMCs through cell death^{117,118}. For example, VSMC uptake of oxLDL and formation of VSMC-derived foam cells has been linked to induction of VSMC apoptosis¹¹⁸, and free cholesterol in the lipid pool might be derived from dead VSMCs¹¹⁹. The microcalcifications (speckles of 0.5–15 μ m) that are sometimes observed within the lipid pools of PITs, typically close to the border with the media, might also be a consequence of VSMC apoptosis⁵¹.

Macrophages can be absent from early PITs⁸⁹ but are a defining characteristic of late-stage PITs and are crucial to the progression of PITs to fibroatheromas. Lineage-tracing studies have shown that cells expressing macrophage markers in early lesions in mice (which resemble intimal xanthomas) are mostly derived from recruited circulating monocytes^{23,120} and might also derive from proliferation of resident macrophages^{121,122}. However, the presence of cells co-expressing α SMA and CD68 in human atherosclerotic plaques indicates that VSMCs are also likely to contribute substantially to the population of cells that express macrophage markers in early plaques^{5,61}. Monocytes are recruited to PITs through the expression of adhesion molecules (including selectins, intercellular adhesion molecule 1,

Response to retention hypothesis

Hypothesis that subendothelial retention of lipid, in the form of lipoproteins, is the initial step in atherogenesis.

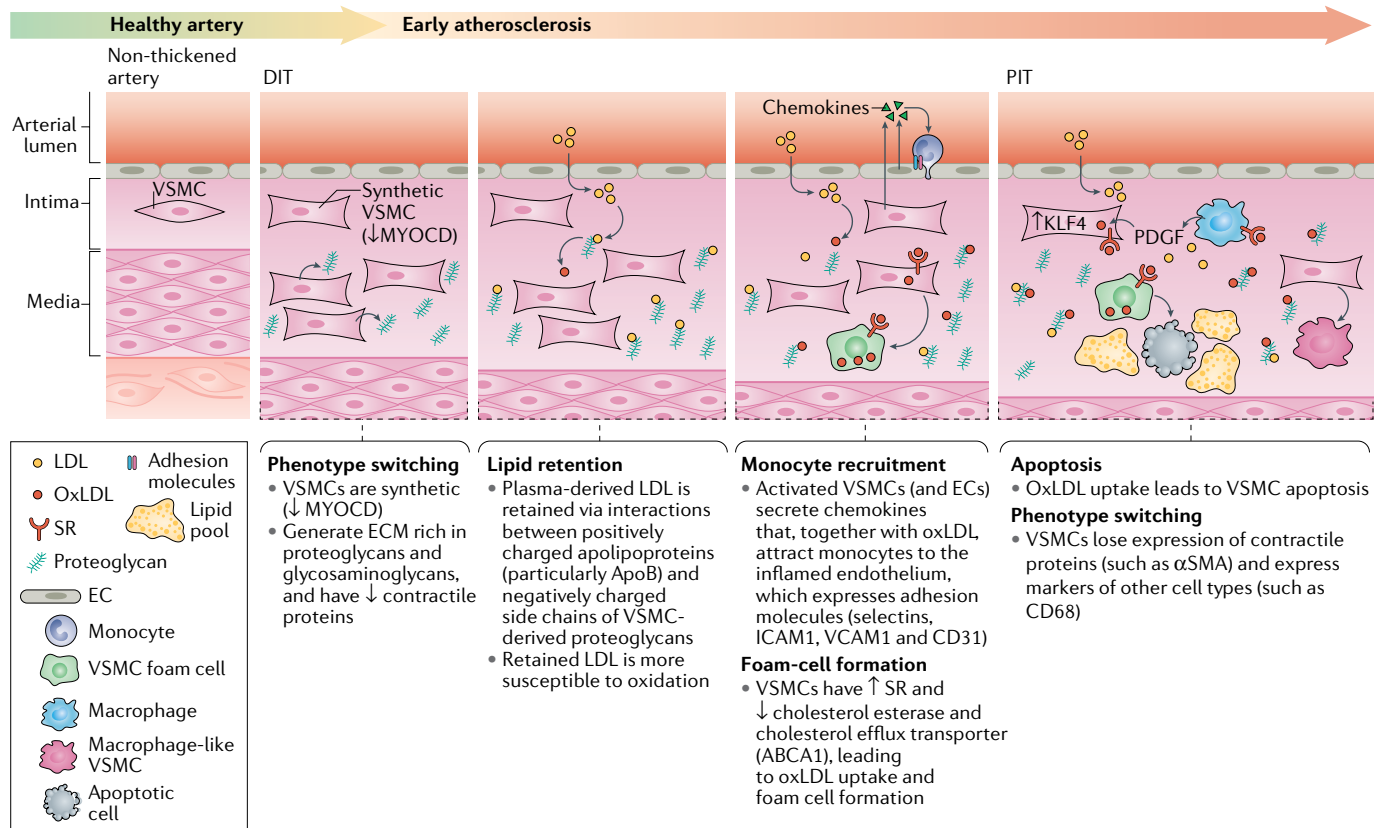


Fig. 2 | VSMCs in early atherosclerosis. Summary of the role of vascular smooth muscle cells (VSMCs) in early atherosclerosis, during the progression from diffuse intimal thickening (DIT) to pathological intimal thickening (PIT). In early atherosclerosis, VSMCs are the predominant cell type within the plaque and are the main source of atherogenic, lipid-retentive extracellular matrix (ECM) rich in proteoglycans (biglycan, versican and perlecan) and glycosaminoglycans (hyaluronan). LDL from the circulation is retained in the arterial intima through the interaction of LDL apolipoproteins with VSMC-derived proteoglycans (chondroitin sulfate in biglycan and versican and heparin sulfate in perlecan). The retained LDL is susceptible to modifications such as oxidation (oxLDL). Uptake of oxLDL by VSMCs leads to foam cell formation and apoptosis. Activated VSMCs secrete chemokines, such as CC-chemokine ligand 2 (CCL2), CCL5 and CXC-chemokine ligand 1 (CXCL1), and contribute to the recruitment of monocytes, which differentiate to macrophages. Progression to PIT is typically associated with a decreased number of cells with VSMC markers (such as α-smooth muscle actin (αSMA)⁺) and increased number of cells with macrophage markers (such as CD68⁺), probably reflecting a combination of VSMC death and VSMC phenotype switching to macrophage-like cells (as a consequence of decreased myocardin (MYOCD) and increased Krüppel-like factor 4 (KLF4) expression). ABCA1, ATP-binding cassette transporter; ApoB, apolipoprotein B; EC, endothelial cell; ICAM1, intercellular adhesion molecule 1; PDGF, platelet-derived growth factor; SR, scavenger receptor; VCAM1, vascular cell adhesion molecule 1.

vascular cell adhesion molecule 1 and CD31) in endothelial cells¹²³ and the presence of chemoattractants, including chemokines (such as CCL2, CCL5 and CXCL1 (REFS^{120,124}), which are secreted by VSMCs and endothelial cells after stimulation with pro-inflammatory cytokines or oxLDL *in vitro*¹²⁵) and modified lipids (such as oxLDL¹²⁶). Studies in animal models have collectively revealed an essential requirement for macrophages in the progression of atherosclerosis^{120,122,127,128}, which is likely to involve effects on VSMC migration, proliferation (through production of factors such as PDGF¹²⁹) and phenotype switching¹³⁰.

Late atherosclerosis. PITs can progress to fibroatheromas (FIG. 3), which are characterized by the presence of a fibrous cap and a necrotic core, the origins of which are the extracellular lipid pools and insufficient efferocytosis of dead VSMCs and macrophages^{131–133}. This phase

of atherosclerosis (late PIT to early fibroatheroma) depends on extensive accumulation of macrophages on the luminal side of the lipid pools, where they phagocytose deposited lipids to become foam cells. In the absence of resolution, the ensuing inflammatory reaction is self-perpetuating; macrophages and VSMCs become foam cells, which eventually die, mostly by apoptosis but potentially through other mechanisms (BOX 4). Given that the plaque milieu suppresses efferocytosis^{133–136}, uncleared apoptotic cells subsequently undergo secondary necrosis with release of further inflammatory material, such as damage-associated molecular patterns (DAMPs)¹³⁷. The accompanying healing response involves the formation of the fibrous cap, which, at least in the early stages of atherosclerosis, is a highly cellular region, rich in VSMC-derived αSMA⁺ cells^{22,38–40} among an altered ECM that has decreased proteoglycan levels and an increase in the proportion of collagens (mostly type I and III).

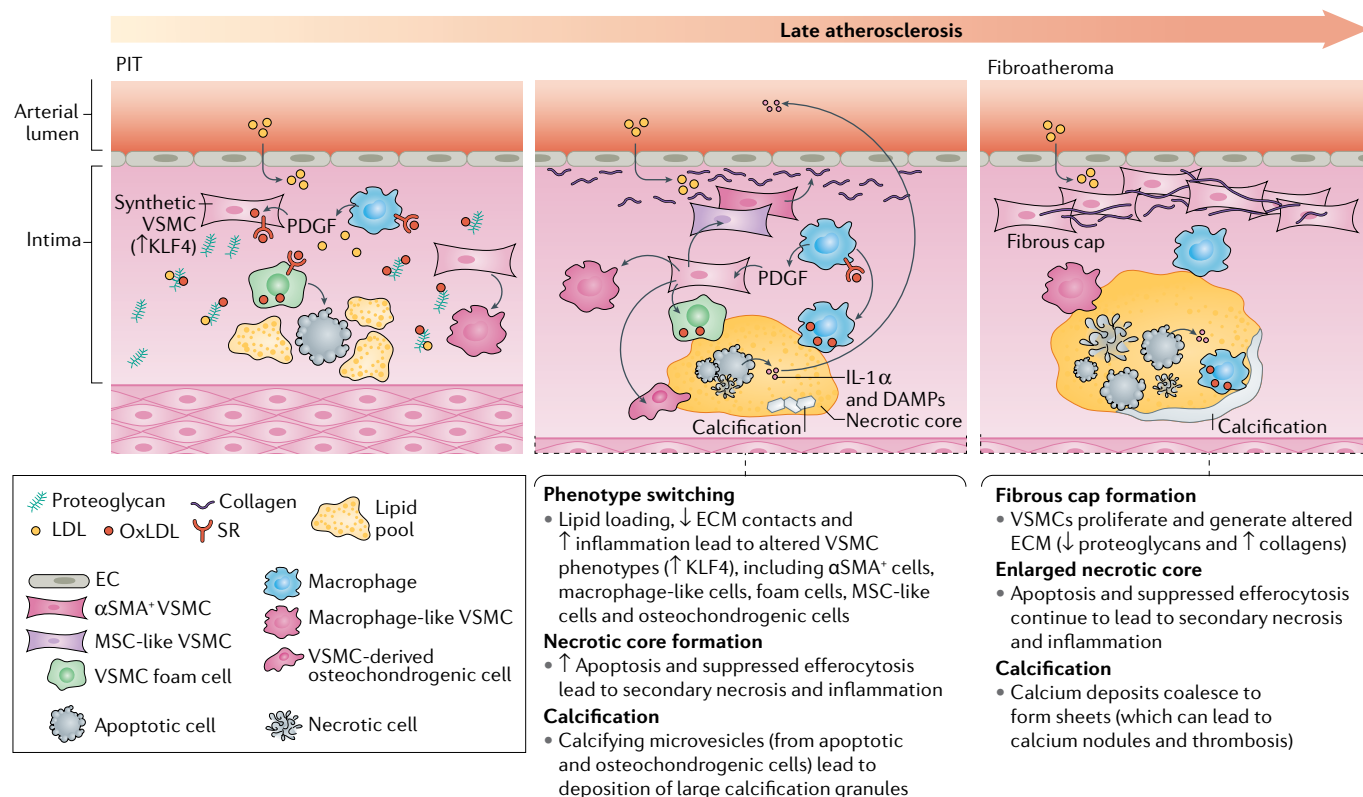


Fig. 3 | VSMCs in late atherosclerosis. Summary of the role of vascular smooth muscle cells (VSMCs) in late atherosclerosis, during the progression from pathological intimal thickening (PIT) to fibroatheroma. This phase of atherosclerosis is characterized by the generation of the fibrous cap by VSMCs and the formation of the necrotic core, which is the consequence of defective efferocytosis of apoptotic cells (mostly VSMCs and macrophages). Through phenotype switching, VSMCs contribute to many cell phenotypes in the plaque, including the extracellular matrix (ECM)-producing cells of the fibrous cap, macrophage-like cells, foam cells, mesenchymal stem cell (MSC)-like cells and osteochondrogenic cells. VSMCs also contribute to calcification through a number of mechanisms, including apoptosis and osteochondrogenic conversion. α SMA, α -smooth muscle actin; DAMPs, damage-associated molecular patterns; EC, endothelial cell; KLF4, Krüppel-like factor 4; oxLDL, oxidized LDL; PDGF, platelet-derived growth factor; SR, scavenger receptor.

In mice, the VSMCs in the fibrous cap are derived from medial VSMCs^{22,138} that have undergone migration and proliferation in response to cytokines and growth factors, such as PDGF derived from macrophages and activated endothelial cells^{23,129,139}. This initial stage of VSMC recruitment depends, at least in part, on the transcription factor OCT4 (REF.²¹). In humans, both pre-existing intimal and medial VSMCs can contribute to plaque VSMCs⁴⁸. Definitive proof that VSMCs are responsible for the production of the ECM of the fibrous cap is lacking. However, this hypothesis is consistent with colocalization of collagen synthesis and VSMCs in the fibrous cap¹⁴⁰, correlation of fibrous cap thickness with VSMC phenotype in mice^{11,21,141} and the correlation of fibrous cap stability with VSMC cell number in humans¹⁴². In addition, a study of VSMC-specific deletion of *Col15a* in mice resulted in a >70% reduction in collagen type XV content in atherosclerotic lesions, supporting VSMCs as the major source of this collagen in atherosclerotic plaques¹⁴³. Further evidence that VSMCs are the major source of collagens in atherosclerotic lesions comes from studies in vitro, including proteomic analysis of the secretome of lipid-loaded VSMCs¹⁰⁹ and the induction of collagen synthesis by

VSMCs in culture by TGF β , PDGF, IL-1, angiotensin II, cholesterol, homocysteine and mechanical stretch^{144,145}.

VSMCs in the later stages of atherosclerosis have previously been thought to have an entirely beneficial role, for example, by stabilizing the plaque through the generation of the fibrous cap. However, lipid loading of VSMCs and altered interactions with the ECM lead to changes in the VSMC phenotype and to increased expression of macrophage markers in VSMCs⁵⁹. Indeed, VSMCs contribute to 30–70% of cells expressing macrophage markers^{11,20} and to a similar percentage of foam cells¹⁴⁶ in mouse atherosclerotic plaques and to approximately 30–40% of CD68⁺ cells and 50% of foam cells in human atherosclerotic plaques^{11,60}. VSMC-specific deletion of the gene encoding the transcription factor KLF4 reduces VSMC switching to macrophage-marker-positive cells and results in a marked increase in the thickness and α SMA⁺ cell content of the fibrous cap¹¹. Although these studies have shown that VSMCs can express macrophage markers, in vitro studies of the transcriptomes of both mouse and human VSMC-derived foam cells and macrophage-derived foam cells indicate that these cell types are functionally distinct and that VSMC-derived foam cells have reduced

Vulnerable plaque

Atherosclerotic plaque with a phenotype associated with increased risk of rupture, also known as thin-cap fibroatheromas, defined by a thin fibrous cap (<65 µm) and a large necrotic core.

phagocytic and efferocytic responses compared with macrophage-derived foam cells⁵⁹. VSMCs have long been known to contribute to the inflammatory milieu of the atherosclerotic plaque through recruitment of macrophages; however, these studies strongly suggest that VSMC-derived macrophage-like cells also directly affect plaque progression.

In early fibroatheromas, calcification is observed as large granules in the necrotic core and the surrounding ECM, resulting from a number of interrelated processes, including macrophage-derived and VSMC-derived calcifying microvesicles^{147–149}, release of apoptotic bodies¹⁵⁰ or the activity of osteochondrogenic cells¹⁵¹. As the fibroatheroma develops, the microcalcifications can coalesce into larger speckles and fragments that can form sheets or plates¹⁴⁹ that are visible by tomography. Fragmentation of these sheets and fibrin encapsulation can lead to the formation of calcium nodules, which

protrude into the vessel lumen and precipitate thrombosis⁸⁸. The extent of plaque calcification varies according to the vascular bed, and studies have linked this variation to the different propensities of the local, developmentally distinct VSMCs to promote calcification^{152,153}. VSMCs have long been linked to calcification^{150,154}, and osteochondrogenic conversion of VSMCs in vitro is increased by environmental cues found in atherosclerotic plaques, including phenotypic conversion¹⁵⁵, apoptotic bodies¹⁵⁰, oxLDL¹⁵⁶ and pro-inflammatory cytokines such as tumour necrosis factor (TNF)¹⁵⁷, IL-1β¹⁵⁸ and IL-18 (REF.¹⁵⁹). Furthermore, specific genetic modulation of VSMC osteochondrogenesis in vivo leads to altered calcification in animal models of atherosclerosis^{160–162}. Most convincingly, however, one study established that most of the osteochondrogenic precursor (Runx-related transcription factor 2 (RUNX2)⁺) cells and chondrocyte-like (type II collagen positive) cells of mouse atherosclerotic plaques derive from VSMCs¹³⁸.

Box 4 | Mechanisms of cell death

Apoptosis

Apoptosis is the most common form of programmed cell death (PCD) throughout development and day-to-day physiology. Apoptosis is executed by pro-apoptotic caspases (such as caspase 3 and caspase 7), with main initiation pathways controlled via the mitochondria (via BCL-2 family members) or via death receptors in the cell membrane (for example, tumour necrosis factor (TNF) receptor 1 and other TNF receptor superfamily members such as FAS). Apoptotic cells must be phagocytosed; otherwise, secondary necrosis with leakage of pro-inflammatory contents (including damage-associated molecular patterns) occurs. All major cell types within the atherosclerotic plaque have been shown to undergo apoptosis.

Autophagic cell death

Autophagy is a mechanism for the organized degradation and recycling of intracellular components within double-membraned autophagosomes that fuse with lysosomes and can be a response to stress that enables the cell to survive but can also occur as PCD. Vascular smooth muscle cell (VSMC)-specific deficiency in autophagy leads to increased VSMC death and increased features of vulnerable atherosclerotic plaques¹⁸⁸.

Necrosis

Necrosis is a non-programmed form of cell death characterized by catastrophic loss of plasma membrane integrity and leakage of cell contents. Apoptotic cells that are not cleared by phagocytes default to secondary necrosis. This type of cell death is difficult to identify in vivo, but ultrastructural evidence suggests that macrophage and VSMC necrosis occurs in atherosclerotic plaques.

Necroptosis

Necroptosis is a programmed form of necrosis that enables cell suicide when apoptosis is blocked (such as by viral caspase inhibitors). Necroptosis is mediated by receptor-interacting serine/threonine-protein kinase 1 (RIPK1)–RIPK3, which form the ripoptosome that activates mixed lineage kinase domain-like protein (MLKL), which destroys the plasma membrane. Increased RIPK3 and MLKL levels have been reported in human atherosclerotic plaques, but necroptosis is difficult to detect in vivo.

Pyroptosis

Pyroptosis is an inflammatory form of cell death that occurs in concert with inflammasome activation and IL-1 production, often in response to intracellular infection. Pyroptosis leads to the activation of pro-inflammatory caspases (such as caspase 1, caspase 4, caspase 5 and caspase 11), which activate IL-1 and/or the pore-forming protein gasdermin D (GSDMD), and subsequent membrane permeabilization. Pyroptosis probably occurs in atherosclerotic plaques after the activation of macrophage NLRP3 inflammasomes by cholesterol crystals.

Paraptosis

Paraptosis refers to a caspase-independent mechanism of cell death that leads to cytoplasmic vacuolation and eventual osmotic lysis. To date, this type of cell death has not been described in atherosclerotic plaques.

Clinical sequelae. The major clinical sequelae of atherosclerosis depend on the anatomical site of the vascular bed involved (angina and myocardial infarction in coronary arteries and stroke in carotid arteries) and typically manifest as a result of thrombosis. The primary cause (accounting for ~60–70% of cases) of thrombosis is plaque rupture¹⁶³, and the remaining cases are predominantly the result of plaque erosion (the latter being much more frequent in young individuals, particularly women) (FIG. 4). A minority (typically ~5%) are due to thrombosis forming on calcified nodules. However, thrombosis and clinical sequelae are not an inevitable consequence of atherosclerosis; analysis of autopsies has revealed that atherosclerotic plaques often show evidence of silent (non-occlusive) thrombi that have undergone repair and healing. Furthermore, the widespread implementation of clinical interventions, including lipid-lowering therapies, is changing the clinical presentation of atherosclerosis, in association with changes in the characteristics of the ‘vulnerable plaque’²¹⁶⁴.

As the fibroatheroma develops, so does the necrotic core; the free cholesterol content and calcification increases, and breakdown and remodelling of the fibrous cap ECM occur. The latter is thought to be principally due to the actions of proteases (in particular MMPs¹⁶⁵) but also to sulfatases and exoglycosidases that are predominantly released by macrophages¹⁶⁶ but can also come from VSMCs¹⁶⁷. Concomitantly, VSMCs are depleted from the fibrous cap through cell death, and therefore the cap diminishes, while the growing necrotic core extends outwards, leading to thinning of the fibrous cap^{168,169}. Thin-cap fibroatheromas are defined by a fibrous cap of <65 µm and are also known as ‘vulnerable plaques’ because studies have shown that these plaques are highly susceptible to rupture. The underlying mechanisms of plaque rupture are ill-defined but are thought to be associated with proteolytic activity^{166,167}, mechanical stress¹⁷⁰ and microcalcification of the fibrous cap^{149,171}.

Plaque rupture is inversely correlated with VSMC number¹⁴², which is determined by proliferation,

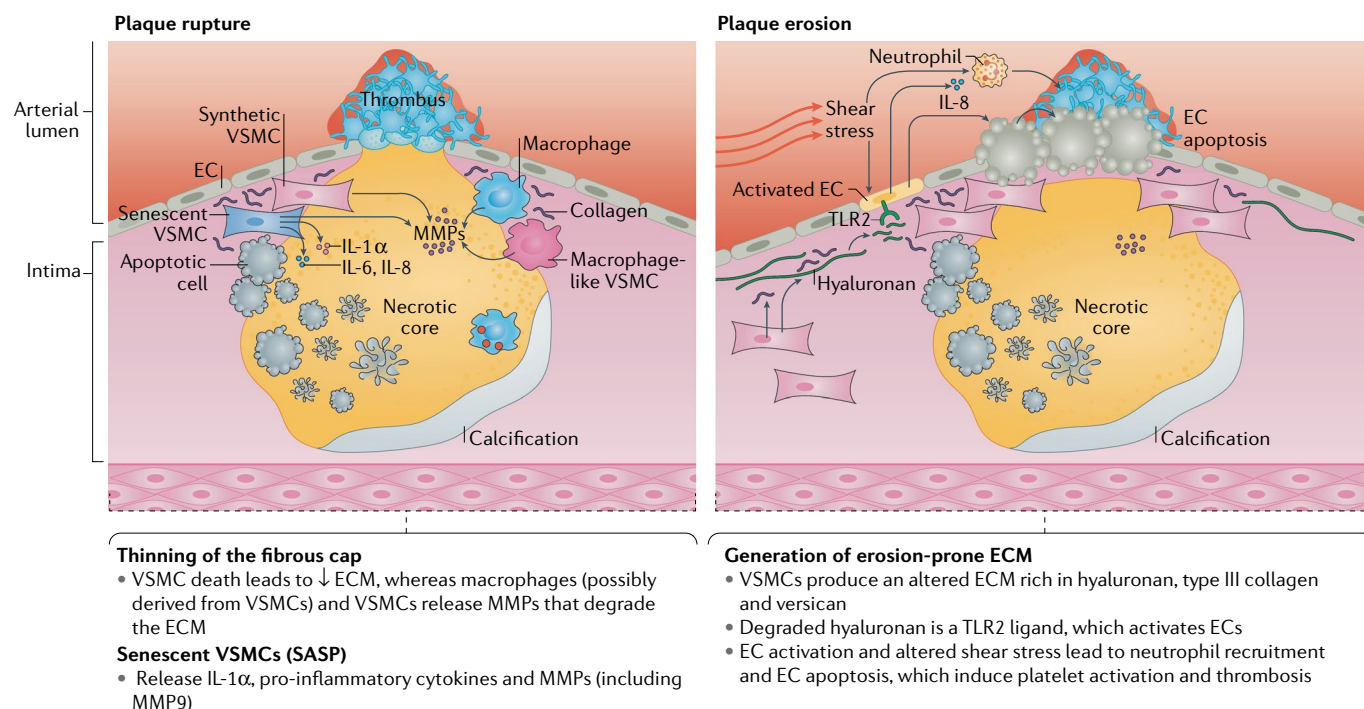


Fig. 4 | **VSMCs in clinical sequelae of atherosclerosis.** Summary of the role of vascular smooth muscle cells (VSMCs) in atherosclerotic plaque rupture and plaque erosion, the two major processes underlying thrombosis and, thereby, the clinical sequelae of atherosclerosis. EC, endothelial cell; ECM, extracellular matrix; MMP, matrix metalloproteinase; SASP, senescence-associated secretory phenotype; TLR, Toll-like receptor.

migration and death of VSMCs. Advanced atherosclerotic lesions in humans show little VSMC proliferation^{172,173}, but VSMC death, through apoptosis and necrosis (BOX 4), is increased compared with normal vessels^{174–176} and in unstable versus stable plaques¹⁷⁷. Indeed, VSMC apoptosis has been postulated to be crucial to plaque instability¹⁷⁸. Seminal work showed that VSMCs from human atherosclerotic plaques spontaneously undergo apoptosis in vitro, with insulin-like growth factor I (IGF1) and PDGF acting as survival factors¹⁷⁹, and with plaque VSMCs showing reduced IGF1 receptor expression compared with medial VSMCs from non-atherosclerotic arteries¹⁸⁰. Similarly, cell-to-cell contact via N-cadherin promotes VSMC survival¹⁸¹. Conversely, numerous factors that induce VSMC apoptosis have been described in atherosclerotic plaques, including cell-directed killing (by macrophages, T lymphocytes and mast cells), reactive oxygen species, DNA damage, anoikis and cholesterol. Studies of VSMC apoptosis in vivo have used mice that have either alterations in apoptotic pathways or systems to induce apoptosis. Early work showed that adenoviral-mediated expression of p53 in atherosclerotic plaques in mice led to VSMC apoptosis and cap thinning¹⁸². Similarly, diphtheria-toxin-mediated induction of apoptosis specifically in VSMC revealed that short-term VSMC apoptosis within established atherosclerotic plaques had no effect on plaque size but resulted in vulnerable plaques with small fibrous caps and a paucity of VSMCs and structural matrix compared with those in control mice¹⁷⁸, a finding that was subsequently corroborated by many studies in which VSMC

death was promoted or inhibited^{167,181,183–187}. Strikingly, diphtheria-toxin-mediated induction of VSMC apoptosis alongside high-fat feeding during atherogenesis in mice results in larger atherosclerotic plaques⁵¹, showing that the consequences of VSMC death are more than cell loss alone, and in fact actively drives plaque growth — another well-replicated finding^{167,185,187,188}. A crucial controller of VSMC apoptosis in atherosclerotic plaques in vivo seems to be the survival kinase AKT1 (REFS^{183,186,187}). Conditional ablation of *Akt1* during atherogenesis in mice induces VSMC apoptosis and larger plaques, and *Akt1* ablation in established plaques leads to a reduced fibrous cap. The contribution of VSMC death to plaque stability is complex and extends beyond direct cell loss, with further consequences on the local milieu (such as the initiation of calcification¹⁵⁰) and wider effects in activating the immune system. The atherosclerotic plaque environment is known to inhibit phagocytosis^{133–136}, and defective efferocytosis of apoptotic cells, leading to secondary necrosis and leakage of intracellular contents, has been proposed to exacerbate the inflammatory milieu^{131,132,137}. Indeed, necrotic VSMCs potentially drive inflammation via IL-1 α release owing to a lack of IL-1 receptor type 2 in VSMCs, which normally binds to and inhibits IL-1 α ^{133,189}. Therefore, a consensus appears whereby functional VSMCs are essential to maintain the fibrous cap and thereby plaque stability, but death of VSMCs is a potent driver of atherogenesis.

A study of the VSMC transcriptome in symptomatic versus asymptomatic carotid atherosclerotic plaques has also highlighted the importance of VSMC senescence in

promoting plaque instability¹⁹⁰. Unstable, mature plaques show low VSMC proliferation and clear evidence of VSMC senescence¹⁹¹. Senescent VSMCs were originally thought to promote plaque instability through inaction; that is, a lack of VSMC proliferation and matrix production leads to weakening of the fibrous cap. However, senescent VSMCs establish a robust IL-1 α -driven SASP comprising multiple inflammatory cytokines, chemokines, MMPs and osteogenic factors^{80,192}. Thus, the VSMC SASP can recruit mononuclear cells, induce adhesion receptor expression in endothelial cells and activate adjacent normal VSMCs⁸⁰, effectively amplifying the effect of a small number of senescent VSMCs. Senescent VSMCs also produce less collagen and release active MMP9 (REF.⁸⁰), whereas BMP2 and osteoprotegerin associated with the SASP drive calcification¹⁹². Thus, senescent VSMCs can have a negative influence on atherosclerotic plaques through both loss of normal function and a direct effect on the local milieu.

An alternative route to thrombosis and clinical sequelae is through plaque erosion. Erosion refers to the formation of a thrombus in the absence of the atherosclerotic plaque rupture at sites of endothelial denudation or disruption. The underlying atherosclerotic plaque can be an intimal thickening or fibroatheroma^{88,169}, but VSMCs are often abundant, amidst a proteoglycan-rich ECM, enriched in type III collagen, versican and hyaluronan¹⁹³. Studies have identified an important role for hyaluronan, which upon degradation activates Toll-like receptor 2 signalling¹⁹⁴, and this signalling, combined with altered shear stress, leads to endothelial cell activation and apoptosis¹⁹⁵, neutrophil recruitment and thrombosis¹⁹⁴. Therefore, VSMCs are implicated in the events leading to plaque erosion, in particular as the major source of hyaluronan¹⁹⁶.

Future perspectives

Difficulties in extrapolating studies from mice to humans. Reconciling the results of studies of animal models with those of human atherosclerosis can be challenging, as some important differences exist in how the disease progresses in humans and animal models. This difference is exemplified in the case of DITs, which are absent in most animal models. Another fundamental difference is that fibroatheromas rarely progress to rupture in animal models, exemplified by the effects of an IL-1 β -neutralizing antibody, which were deleterious on the fibrous cap in mice¹⁴¹ but beneficial in reducing cardiovascular events in the CANTOS trial in humans¹⁹⁷. Nonetheless, animal models have been instructive in delineating important pathways and basic principles that might underlie plaque development in humans. This is particularly true of the lineage-tracing studies in mouse models of atherosclerosis, which have unambiguously established the importance of clonality and phenotype switching of VSMCs. Combinatorial genetic-depletion models are likely to be instrumental in assessing whether biasing the phenotype of VSMC-derived cells in atherosclerosis could be a potential treatment avenue. Techniques such as mass cytometry (CyTOF) and single-cell omics (genomics, transcriptomics and epigenomics) hold great

promise for high-resolution, spatiotemporal analysis of atherosclerotic plaque cells in situ and are likely to provide the conclusive counterpart data in humans for the aforementioned studies.

VSMCs and genetics of atherosclerosis. In total, >150 loci associated with risk of coronary artery disease (CAD) have been identified from genome-wide association studies and other genetic association studies¹⁹⁸, many of which are associated with the disease independently of other known risk factors. Therefore, elucidation of the underlying molecular mechanisms of atherosclerosis might reveal novel pathways and hence targets for therapeutic intervention. However, identification of causal variants is usually far from trivial; CAD risk loci are often located in non-coding regions, where the causal variant is predicted to affect regulation of gene expression, which might operate over large distances and be cell-type or context specific. Studies are ongoing to identify and functionally characterize the causal variants in each of the CAD risk loci, and in vitro studies of VSMCs are proving an invaluable resource in this quest. Integration of transcriptomic and epigenomic maps from VSMCs (and other atherosclerotic plaque cells) with those of the genetic architecture of CAD can be very informative for prioritizing variants (and potential pathways) for functional characterization^{199,200}. Unsurprisingly, given the crucial role of VSMCs in atherosclerosis, a number of loci have been predicted to modulate the risk of CAD through mechanisms specific to VSMCs²⁰⁰. Thus, studies in cultured VSMCs and VSMCs derived from stem cells^{49,201} are likely to be instrumental in the functional characterization of CAD variants. Pioneering examples of such studies include the characterization of the *SMAD3* and *TCF21* loci²⁰².

Conclusions

The role of VSMCs in atherosclerosis extends far beyond that perceived for decades. VSMCs and VSMC-derived cells comprise a (if not the) major source of atherosclerotic plaque cells and contribute to numerous plaque cell phenotypes, including macrophage-like cells and foam cells, in addition to cells responsible for producing the atherogenic or athero-protective ECM throughout the disease process. Thus, VSMCs are implicated mechanistically at all stages of atherosclerosis, and lineage-tracing studies have established the extent and importance of VSMC clonality and phenotype switching in plaque progression. These concepts have been around for decades, but only very recently technologies for genetic engineering and imaging have converged with a deeper understanding of developmental processes to generate conclusive data in animal models. The era of single-cell omics promises to deliver the evidence as to whether and how these processes contribute to the disease in humans. A better understanding of the biology of VSMCs is required if we are to fulfil aspirations of selectively targeting 'culprit' cells or manipulating cell phenotypes for clinical benefit and/or to avert processes that are detrimental in atherosclerosis.

Published online 26 June 2019

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Author contributions

G.L.B., H.F.J. and M.C.H.C. wrote the manuscript. H.F.J. and M.C.H.C. contributed equally. All the authors researched data for the article, discussed its content, reviewed the manuscript for important intellectual content and edited the manuscript before submission.

Competing interests

The authors declare no competing interests.

Peer review information

Nature Reviews Cardiology thanks D. M. Greif, P. Libby, M. W. Majesky and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

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